

Chameleon – USIA

Sprint 1 – EDA Summary Team: Team B Dataset: On-street Parking Bays

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Team: Team B

Stream: Data Visualization

Research Question: How does parking use change after bike lanes are constructed?

Scope Note

This EDA profiles the static On-street Parking Bays location dataset. It provides the spatial backbone (coordinates + road-segment grouping) for every parking bay in the City of Melbourne. Later, I will integrate this static location data with temporal sensor event data (from 2012- 2020) and the treatment/control street list to answer the core research question. This document will be updated once spatial joins to the study streets are completed.

Overview

The dataset contains 29,053 rows and 7 columns. Each row represents a single on-street parking bay. Bays are grouped by RoadSegmentID.

Metric	Value
Total bays	29,053
Unique road segments	1,413
Unique road segment descriptions	1,412
Bays with KerbsideID	5,072 (17.5%)
Date range of LastUpdated	2025-06-03 to 2026-03-31

Column Summary

Column	Type	Description	Nulls
RoadSegmentID	int	Identifier for the road segment the bay sits on (1,413 unique values)	0%
KerbsideID	string	Identifier that can link to live parking sensor datasets	82.5%
RoadSegmentDescription	string	Human-readable street name + cross-street bounds (near 1:1 with RoadSegmentID)	0%
Latitude	float	Bay latitude	0%
Longitude	float	Bay longitude	0%
LastUpdated	date	Date the bay record was last refreshed (47 distinct dates)	0%
Location	string	Redundant lat/lon pair stored as text	0%

Data Quality Findings

1. KerbsideID is not stable primary key.

- Coverage: Only 17.5% of rows have a KerbsideID, making it unsuitable for linking the majority of bays to sensor data.
- Uniqueness: The KerbsideID is not a reliable primary key. The KerbsideID values are associated with more than one RoadSegmentID (for example: ID 17212 appears on both Berkeley Street and Wellington Parade South).

2. RoadSegmentID is the dimension Key

- The mapping between RoadSegmentID and RoadSegmentDescription is functionally 1:1 (1,413 vs. 1,412 unique values). This makes RoadSegmentID the most reliable and cleanest field for aggregating bays to the street-segment level.

3. LastUpdated is a Static Metadata

- The LastUpdated field has no meaningful time distribution; 64% of rows have the date 2025-09-03, and 25% have 2025-06-03.
- It records the date of snapshot, not a repeated observation over time, is unsuitable for any before or after or time-series analysis.

4. No Duplicate Records

Join Compatibility Assessment

- Attribute Join: The dataset does not contain a `street_segment_id` and cannot be directly joined to the IV treatment/control street list (`street_spatial.gpkg`) via a simple attribute key.
- Spatial Join: The dataset is suit for a spatial join to the IV street list using the Latitude and Longitude columns. This is the most appropriate method to determine which bays fall within the treatment and control areas and is consistent with the project's existing spatial pipeline.

Suitability for Before/After Comparison

- This dataset is not suitable for performing a before/after or difference-in-differences analysis. It is a static, single-moment snapshot with no time-series component.
- Structural Value: Its primary value lies in providing the structural components required for the analysis:
- It has provided precise spatial coordinates (Latitude/Longitude) for every bay.
- It can be used to locate which bays fall inside the IV treatment and control streets once a spatial join is performed.
- The actual time-series signal will come from the sensor event datasets (2012 baseline + later years) joined onto these bay locations.

Next Steps

- Spatially join this dataset to the IV treatment/control street list (via coordinates or `street_spatial.gpkg`) to retain only bays that fall inside the study streets.
- Feed the RoadSegmentID + coordinates into the spatial join pipeline as an additional bridging source between bay locations and the IV street segments.